GREAT LAKES AQUATIC NONINDIGENOUS SPECIES INFORMATION SYSTEM

Invasive Species Threats Assessment

The Restoration of Cisco in Lake Erie

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NOAA

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GLERL 50 YEARS

INTRODUCTION

This report summarizes the threats aquatic invasive species (AIS) pose to the restoration of cisco *Coregonus artedi* in Lake Erie. A qualitative analysis of the Great Lakes Aquatic Nonindigenous Species Information System (GLANSIS) impact database provides insight into the expected impacts of AIS on cisco in Lake Erie and identifies species of greatest concern. Furthermore, a geospatial analysis of GLANSIS distribution data incorporates a geographic component to inform management of regions where invasive species and cisco are expected to co-occur.

Written for the expert panel workshop to assess the threats to the restoration of cisco in Lake Erie, in support of the Coregonine Restoration Framework.

MATERIALS & METHODS

Invasive Species Impacts on Cisco

To examine the impacts of invasive species on cisco, a summary of documented impacts from the literature was constructed using GLANSIS impact data (Table 1). GLANSIS impact data represents all of the impacts documented by staff during the literature review process, prior to the creation of risk assessments and species profiles. The database was recently updated to include all of the impacts documented in the NOAA Technical Memoranda 161 and 169 series, providing a repository of environmental, socio-economic, and beneficial impact data for AIS in the Great Lakes region. Data was extracted from the database on 04/30/2024 using a Python script designed to scrape, clean, and format impact data, including the addition of detailed descriptions and geographic locations (code available on GitHub). Data table output was manually screened for erroneous impact records not applicable to the analysis and finalized in preparation for qualitative analysis.

AIS & Cisco Co-occurrence

To address the prevalence of AIS and cisco interaction and identify key focus areas, a geospatial analysis was performed utilizing GLANSIS AIS distribution data, Lake Erie Subbasins vector layer, and a cisco suitability raster depicting reproductive habitat (Table 1).

A subset of GLANSIS distribution data was curated using the Lake Erie Subbasin data

layer as a selecting feature, the selection is then exported as a new layer. Within the new layer, a select by attributes query using the scientific names for species with observed impacts (Table 2) was implemented to create a point dataset representing the distribution of AIS with impacts on *Coregonus artedi* in the Lake Erie basin. An optimized hot spot analysis was performed on the dataset to identify statistically significant hot spots of AIS that impact cisco. Resulting layer identifies areas that are a 'Hot Spot with 99% Confidence', representing areas with high densities of AIS that impact cisco.

To incorporate *Coregonus artedi* into the spatial analysis, a raster layer (Cisco_Suitability_Layer) depicting suitable reproductive habitat in Lake Erie was imported and reprojected from GRS_1980_ALbers to WGS 1984 Web Mercator using the nearest neighbor method. Raster values were converted to integers and rescaled using raster calculator (expression: Int("Cisco_Suitability_Layer"*100)). Final result is a raster layer in the same coordinate system as invasive occurrence data with a scale of 0-10 in terms of suitable reproductive habitat for cisco. An intersect analysis is performed using the raster layer and hotspot analysis output to identify areas with suitable habitat that also have high densities of impactful AIS.

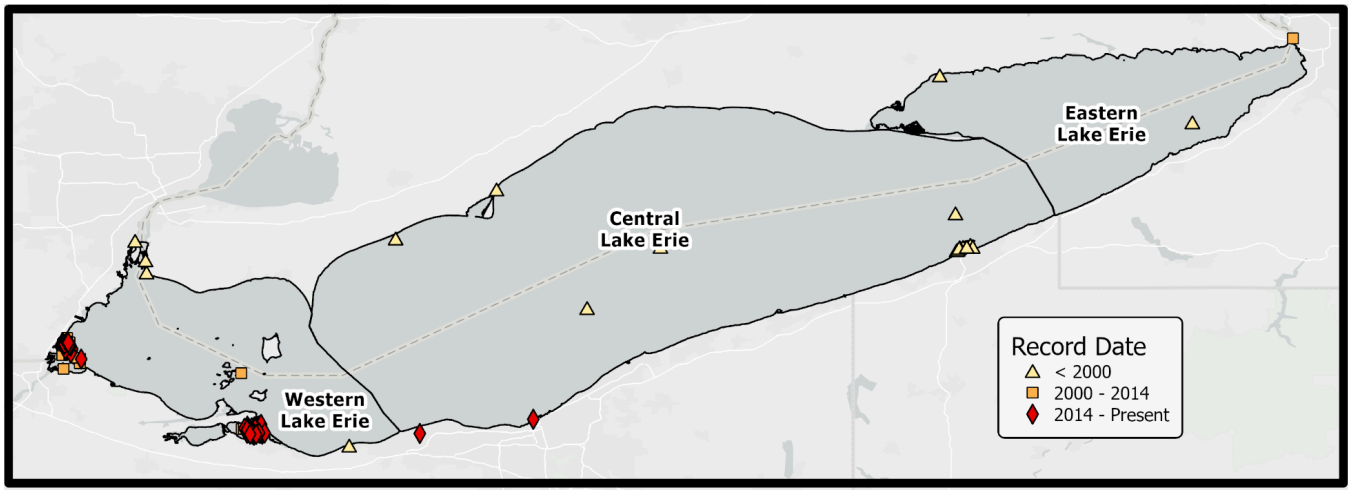
RESULTS

Invasive Species Impacts on Cisco

Extraction of GLANSIS impact data using the Python script resulted in 35 reported impacts from 11 different species, manual screening produced a final dataset of 29 impacts from 10 AIS species. See Table 2 for a list of AIS that impact cisco, identified from GLANSIS Impact Data. The most impactful species is rainbow smelt *Osmerus mordax* with 12 impact records, followed by the spiny waterflea *Bythotrephes longimanus* with 4 records, and a three way tie for third between sea lamprey *Petromyzon marinus*, bacterial kidney disease *Renibacterium salmoninarum*, and alewife *Alosa pseudoharengus* with 2 records each (Figure 1). Distribution data for Lake Erie and a description of impacts on *Coregonus artedi* are presented for each species below.

Rainbow Smelt (*Osmerus mordax*)

Distribution in Lake Erie:



Impact Summary:

Competition

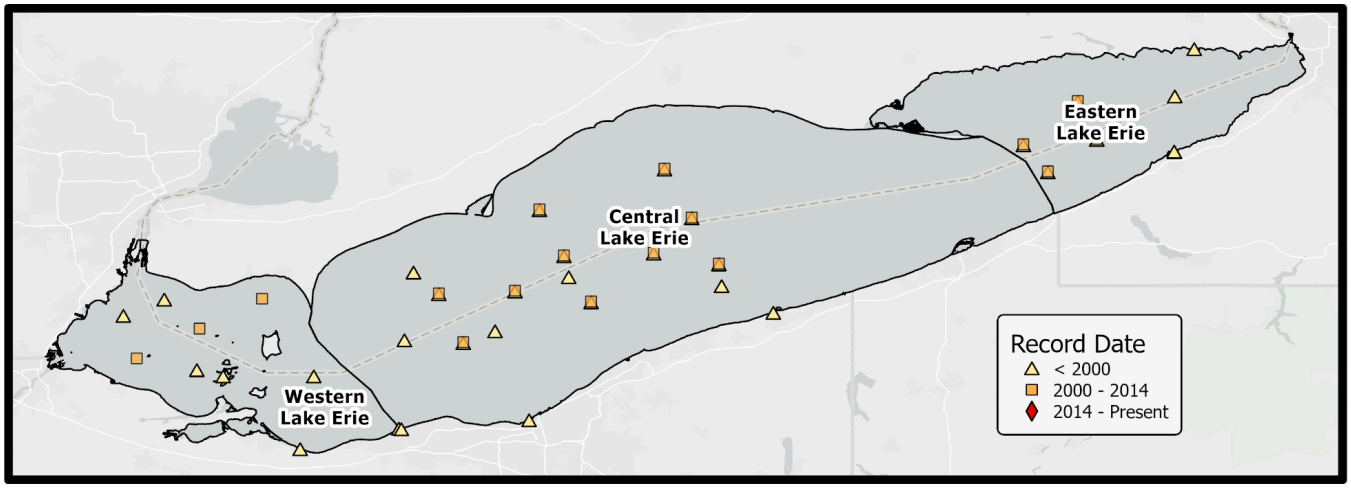
- Rainbow smelt and *Coregonus artedi* have a high probability of habitat overlap, niche overlap in spring-summer diet is high (Rosinski et al. 2020).
- In a review of rainbow smelt introductions in inland Ontario lakes, 5 of 19 lakes with introduced rainbow smelt experienced a decline in cisco (Evans and Loftus 1987).
- Cisco decline was correlated with smelt increases in most of the Great Lakes (Christie 1974).
- Competition with rainbow smelt have been implicated in the declines of shortnose cisco *Coregonus reighardi* and blackfin cisco *Coregonus nigripinnis* (COSEWIC 2005, 2007).

Predation

- Predation by rainbow smelt accounts for 15–52% and 37–100% of the mortality of larval ciscoes in Thunder and Black bays, Ontario (Myers et al. 2009).
- Declines, local extirpations, and limitations to recovery of cisco populations have also been attributed to rainbow smelt predation on larval fish (Stockwell et al. 2009).
- Thermal overlap in Sparkling Lake may have placed young cisco in close proximity to predatory adult smelt during the summer months and indirectly led to the extirpation of cisco from the lake by 1990; only 8 years after smelt were detected (Hrabik et al. 1998).
- Predation by rainbow smelt have been implicated in the declines of shortnose cisco *Coregonus reighardi* and blackfin cisco *Coregonus nigripinnis* (COSEWIC 2005, 2007).

Spiny Waterflea (*Bythotrephes longimanus*)

Distribution in Lake Erie:



Impact Summary:

Competition

- Experimental studies in Harp Lake, Canada suggest that *Bythotrephes longimanus* likely competes with other vertebrate predators for zooplankton, such as *Coregonus artedi* (Dumitru et al. 2001).

Disease/Parasites/Toxicity

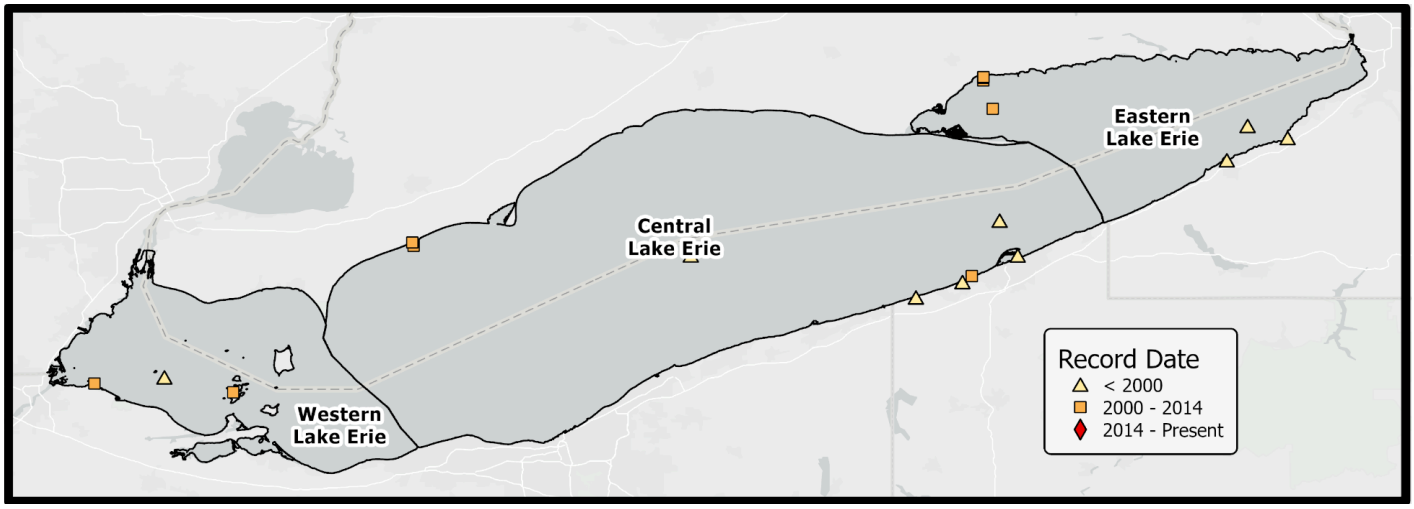
- Significant increases in the trophic position of zooplankton (reduced herbivorous cladoceran biomass and increased omnivorous/predatory copepod biomass) and planktivorous fish such as the lake herring (*Coregonus artedi*) with the introduction of *Bythotrephes longimanus* have the potential to lead to substantial contaminant biomagnification in consumers (Rennie et al. 2011). Increased mercury concentrations in consumers have not been observed following invasion of *B. longimanus;* attributed to increased foraging and growth efficiencies of consumers or to changes in the feeding habits of omnivorous prey.

Beneficial

- *Bythotrephes longimanus* is a food source for fish including yellow perch, white perch, walleye, white bass, alewife, bloater chub, Chinook salmon, emerald shiner, spottail shiner, rainbow smelt, lake herring, lake whitefish, and deepwater sculpin (Branstrator and Lehman 1996).

Sea Lamprey (*Petromyzon marinus*)

Distribution in Lake Erie:



Impact Summary:

Disease/Parasites/Toxicity

- Evidence of impact by predation on *Coregonus* spp., sea lamprey predation led to the extinction of the deepwater cisco (*Coregonus johannae*) and the decline of the blackfin cisco (*C. nigripinnis*), both endemic to the Great Lakes (Miller et al. 1989).
- *Petromyzon marinus* is expected to parasitize *Coregonus artedii* in areas where they co-occur, similar to what has been seen with lake trout in Lake Ontario (Schneider et al. 1996).

Bacterial Kidney Disease (*Renibacterium salmoninarum*)

Distribution in Lake Erie: Not Present in the Lake Erie Basin.

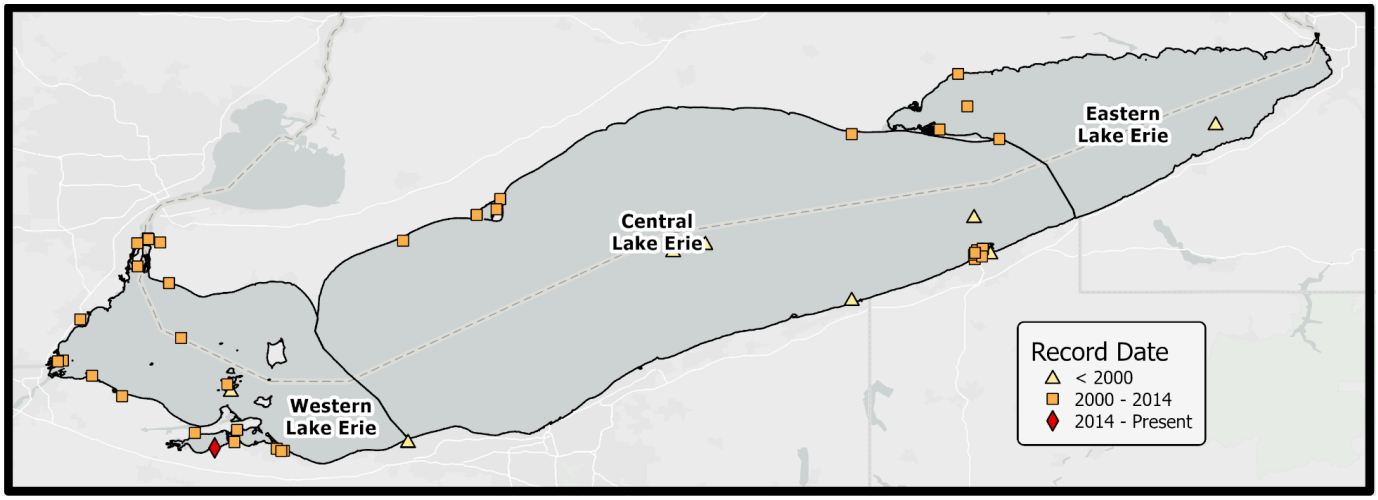
Impact Summary:

Disease/Parasites/Toxicity

- *Coregonus artedi* is a host for *Renibacterium salmoninarum*, many Great Lakes native species have been observed harboring this bacteria (Hay 2003). Data from 2011 and 2015 indicates that *R. salmoninarum* has become more widely distributed with varying prevalence throughout Michigan state hatcheries and wild populations. Detected in cisco at 1.7% to 5% prevalence (GLFHC 2012, 2016).

Alewife (*Alosa pseudoharengus*)

Distribution in Lake Erie:



Impact Summary:

Competition

- Researchers speculated that cisco native to Lake Michigan, shifted habitat and diet as a result of competition with alewife (Crowder 1984).

Genetic

- Crowder (1984) speculated that a cisco native to Lake Michigan, the bloater (*Coregonus hoyi*) evolved fewer and shorter gill rakers as a result of competition with alewife.

Blueback Herring (*Alosa aestivalis*)

Distribution in Lake Erie: Not Present in the Lake Erie Basin, present in Lake Ontario.

Impact Summary:

Competition

- Anecdotal evidence suggests that if blueback herring became established throughout Lake Ontario and/or spread to other Great Lakes it could impede recovery of depressed populations of native fishes such as cisco and lake trout (Owens et al. 1998).

Ruffe (*Gymnocephalus cernua*)

Distribution in Lake Erie: Not Present in the Lake Erie Basin, present in Lake Michigan, Huron, and Superior.

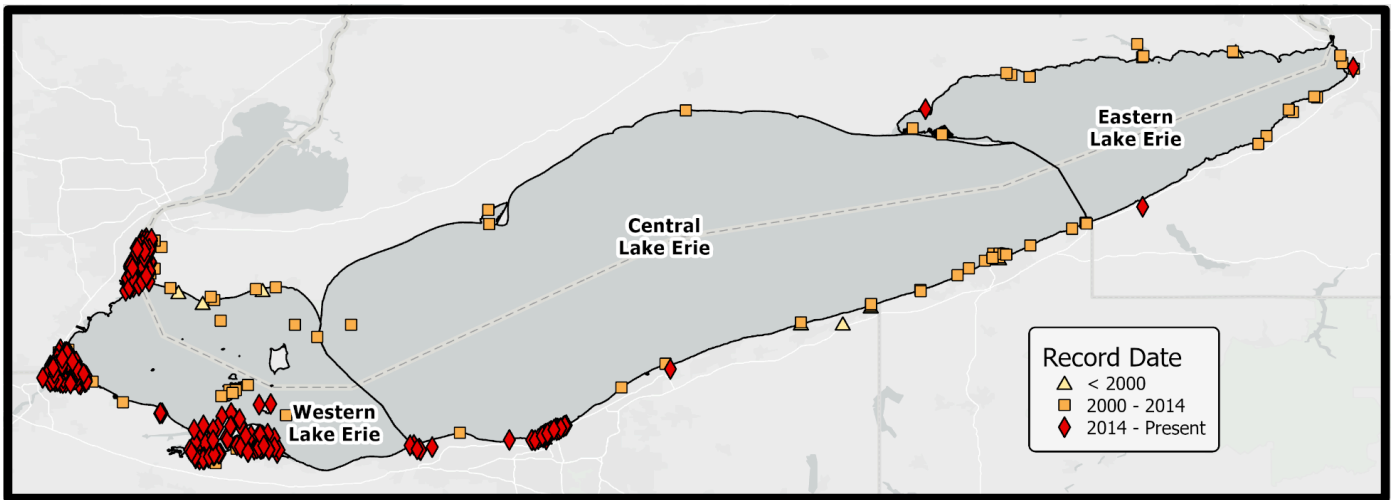
Impact Summary:

Predation

- In Lake Superior, consumption of cisco (*Coregonus artedi*) eggs by ruffe has been documented at a level that could impact the population over winter months (Selgeby 1998).

Round Goby (*Neogobius melanostomus*)

Distribution in Lake Erie:



Impact Summary:

Predation

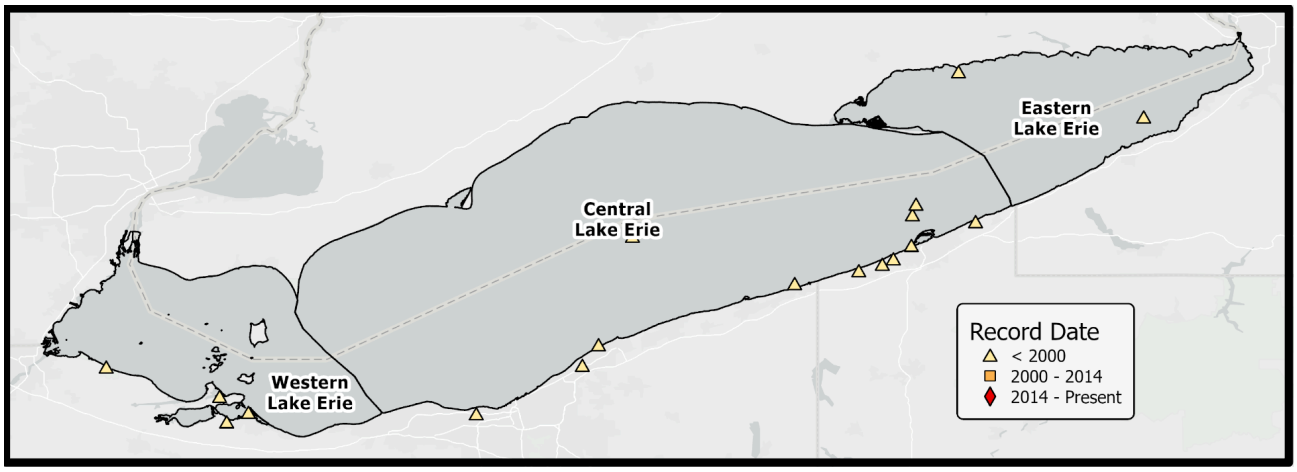
- Recent evidence presented at IAGLR 2024 suggests that *Coregonus artedi* larvae may avoid habitat occupied by round goby. Cisco larvae exhibited tighter group cohesion and frequently avoided round goby in experimental trials (Hoyer et al. 2024).

Beneficial

- Round goby has been incorporated into the diet of *Coregonus artedi* in Lake Ontario (Gatch et al. 2021).

Pink Salmon (*Oncorhynchus gorbuscha*)

Distribution in Lake Erie:



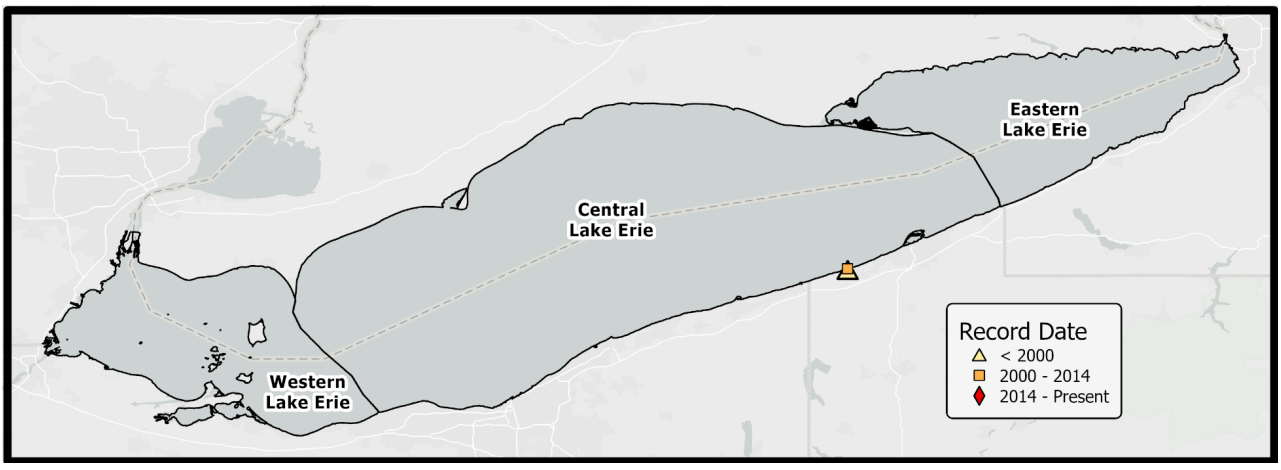
Impact Summary:

Competition

- Pink salmon may displace native cisco by way of food competition (Becker 1983).

Coho Salmon (*Oncorhynchus kisutch*)

Distribution in Lake Erie:



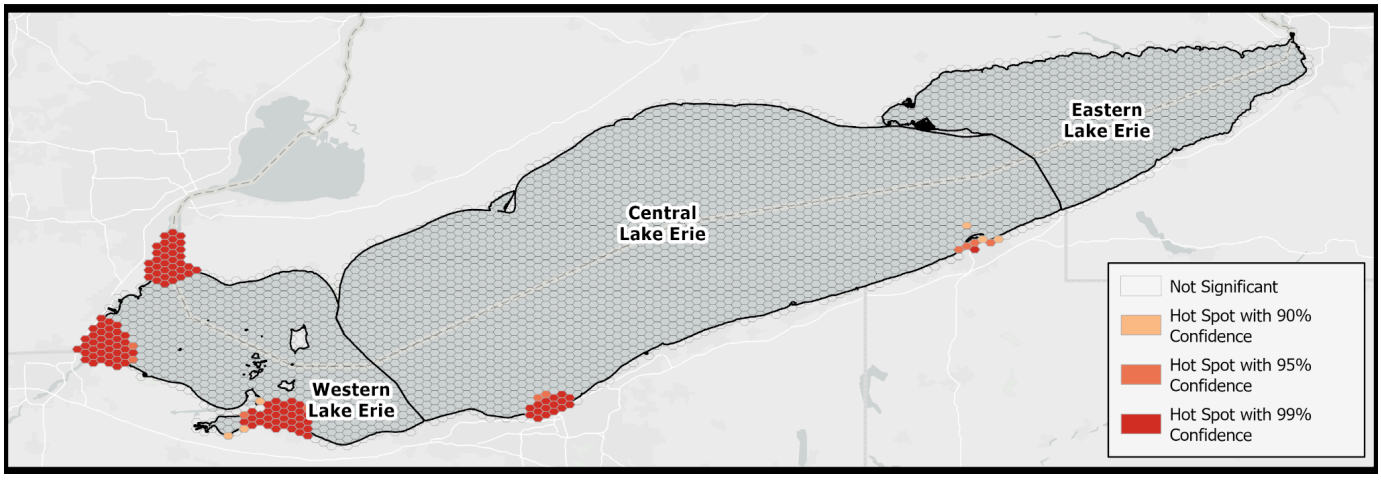
Impact Summary:

Predation

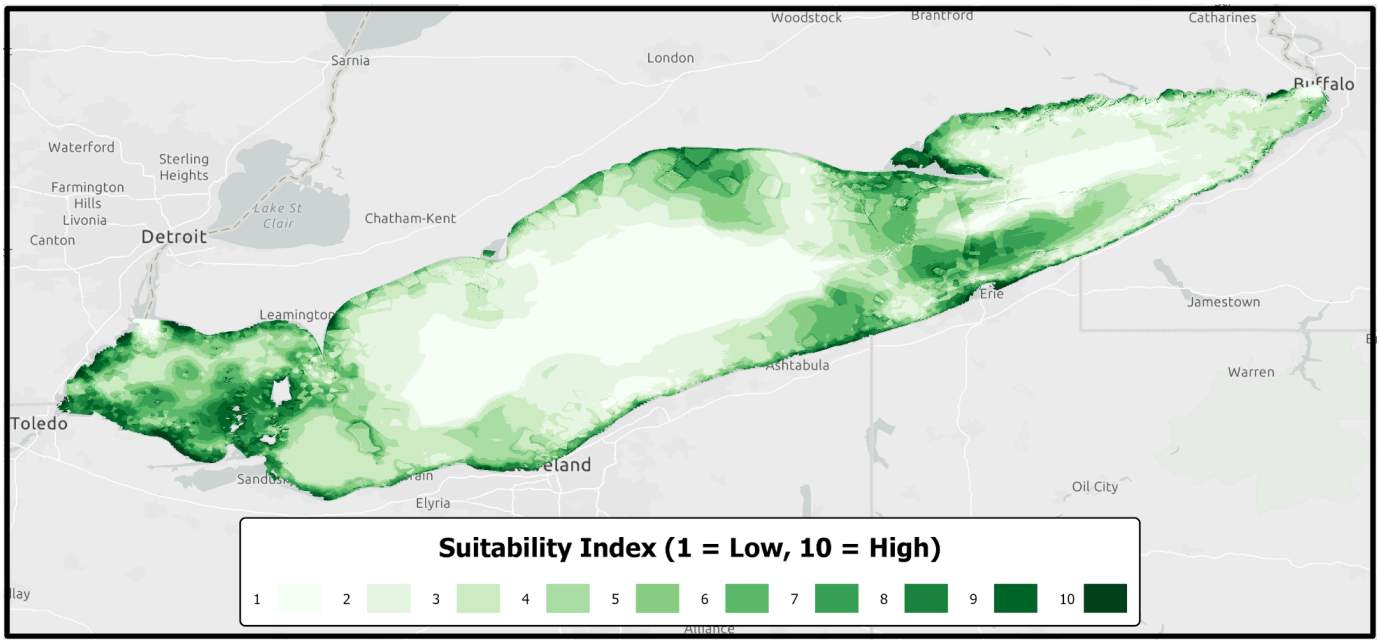
- The diet of coho salmon (Oncorhynchus kisutch) is diverse, including the native cisco *Coregonus artedii* (Crawford 2001).

AIS & Cisco Co-occurrence

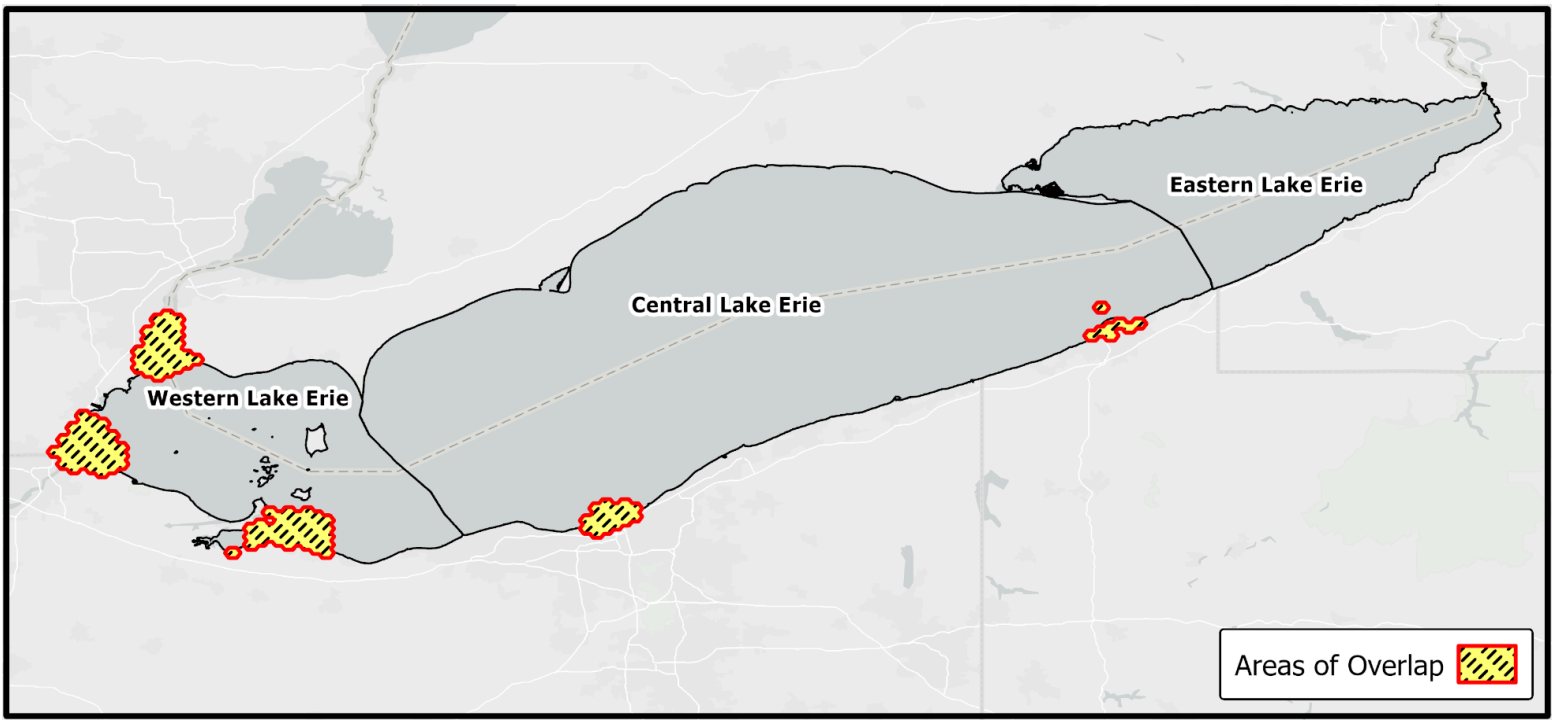
Distribution data for all AIS of interest (Figure 2) was plotted and analyzed using an optimized hot spot analysis, identifying the following hotspots in Lake Erie:



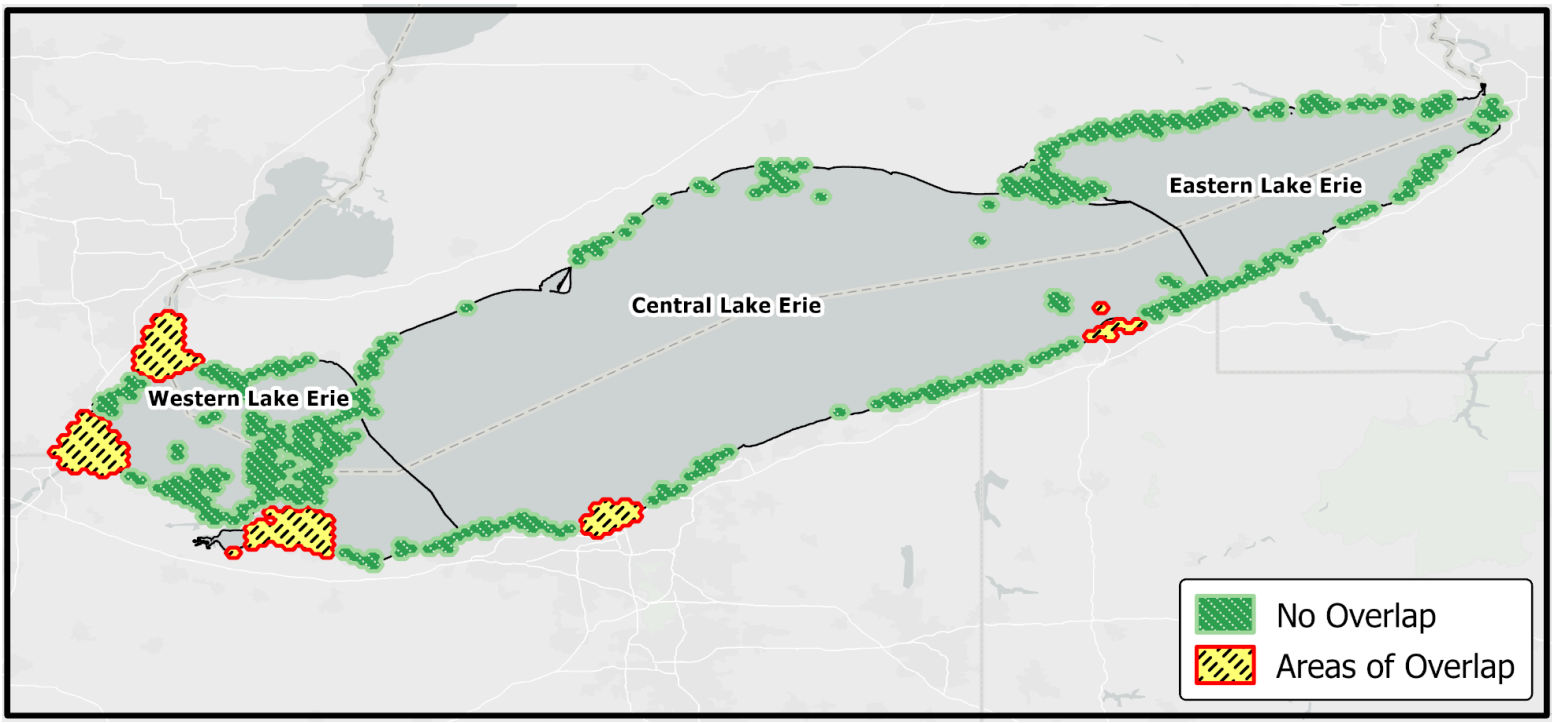
Suitability raster provided by Katelyn King, U of M SEAS (INTERNAL USE ONLY):



Intersect analysis performed on the preceding two layers to extract areas of high suitability (≥ 7.5 on suitability index) that are also hot spots for AIS. The areas of greatest concern are in Western Lake Erie, as seen below:



Alternatively, the inverse can be applied to depict areas of high suitability (≥ 7.5 on suitability index) that are not hot spots for AIS:



The areas of 'No Overlap' represent suitable reproductive habitat for *Coregonus artedi* that is statistically insignificant for the presence of AIS that have documented impacts on *C. artedi* in the literature. Areas of overlap encompass 444.97 square miles whereas the area of no overlap equates to 1708.53 square miles. Results suggest that there is abundant habitat available for *C. artedi* reproduction outside of AIS hot spots. Findings are limited in that conclusions about the distribution of invasive species are drawn from static point records and are unlikely to fully reflect their occupied ranges. Additionally, the suitability raster for *C. artedi* is derived from a MaxEnt model with historical substrate, historical coefficient of variation of ice duration, predicted reef locations, fetch, circulation, and distance to delta wetlands as variables. Historic reproductive habitat may not be an accurate representation of the habitat available today.

CONCLUSION

Qualitative analysis of impacts indicate that *Coregonus artedi* may be susceptible to competition, predation, and parasites as a result of aquatic invasive species present in Lake Erie. Rainbow smelt, broadly distributed in Lake Erie, has been historically correlated with cisco decline in the Great Lakes and has a high probability of competing with *C. artedi* for habitat and food (Christie 1974; Rosinski et al. 2020). The spiny waterflea is expected to compete for zooplankton and has the potential to biomagnify contaminants, but it can also benefit cisco as a food source (Branstrator and Lehman 1996). Sea lamprey parasitism and competition with alewife can impact cisco recovery in Lake Erie while the salmonids are not abundant enough to be considered a significant threat. It is worth noting that the round goby can be incorporated into the diet of *C. artedi*, but recent evidence presented by Hoyer et al. (2024) makes it clear that the impacts of AIS on cisco are not fully understood. Additional research into the impacts of AIS and their mechanisms may be warranted to support the recovery of cisco in Lake Erie.

Tables & Figures

Dataset	Source	Details
Invasive Species Distribution Data	GLANSIS	Publicly available Downloaded: 4/26/2024
Invasive Species Impact Data	GLANSIS	Publicly available on Species Profiles Python tools requires database access Downloaded: 4/30/2024
Cisco Suitability Raster*	Katelyn King, U of M SEAS	Personal Communication MaxEnt model, 5-fold cross validation Received: 5/7/2024
Lake Erie Subbasins	GLAHF	Publicly available Downloaded: 2/7/2023

Table 1. Summary of Data (* = Internal Use Only)

Scientific Name	Common Name	Impact Type
Alosa aestivalis	Blueback Herring	Competition
Alosa pseudoharengus	Alewife	Competition, Genetic
Bythotrephes longimanus	Spiny waterflea	Competition, Diseases/Parasites/Toxicity, Ecoservices*
Gymnocephalus cernua	Ruffe	Predation
Neogobius melanostomus	Round goby	Predation, Food Web*
Oncorhynchus gorbuscha	Pink Salmon	Competition
Oncorhynchus kisutch	Coho Salmon	Predation
Osmerus mordax	Rainbow Smelt	Predation, Competition, Harvest
Petromyzon marinus	Sea Lamprey	Diseases/Parasites/Toxicity
Renibacterium salmoninarum	Bacterial kidney disease (BKD), Dee disease	Diseases/Parasites/Toxicity

Table 2. Species with Impacts on Cisco (* = beneficial impact)

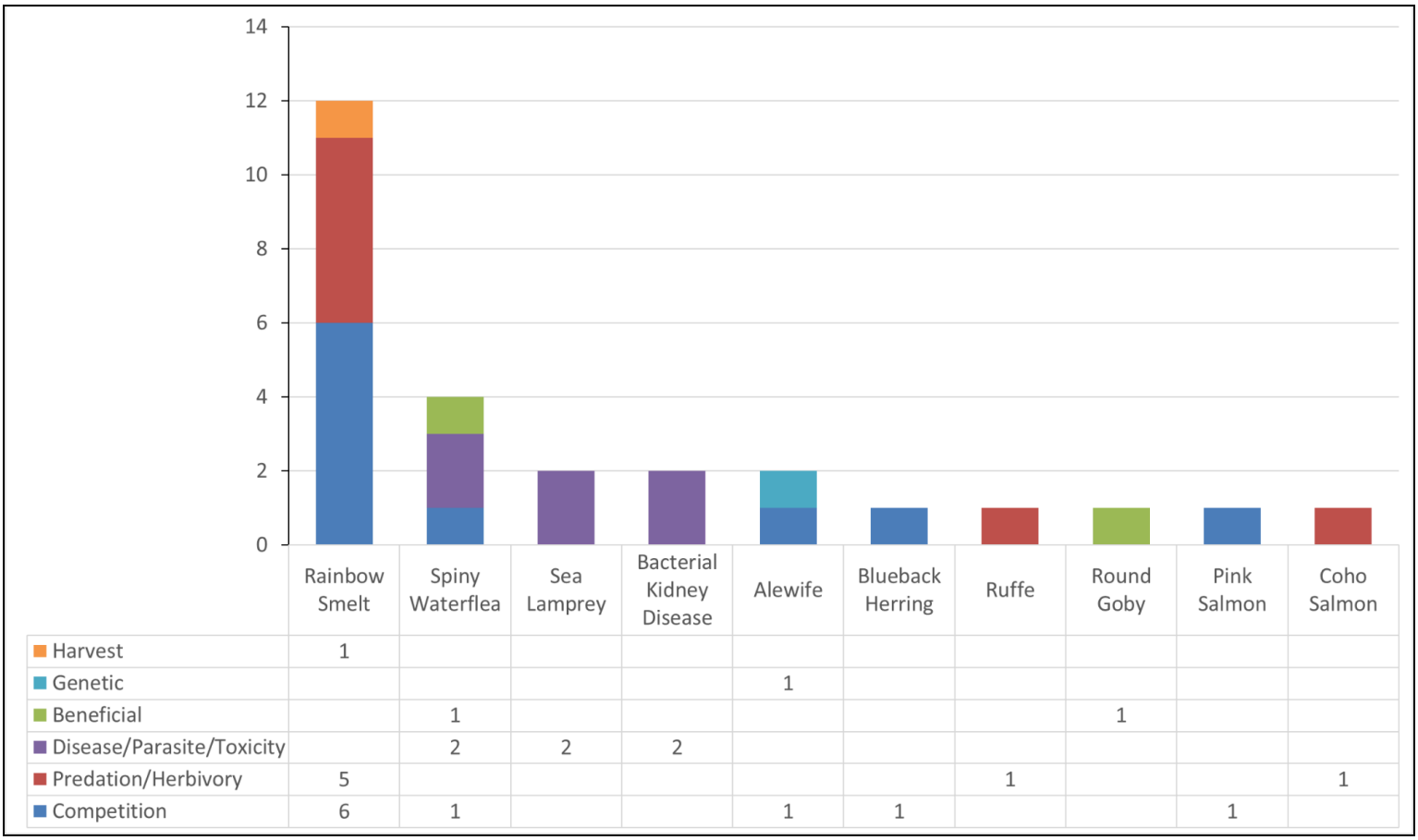


Figure 1. Impact Count by Species and Category

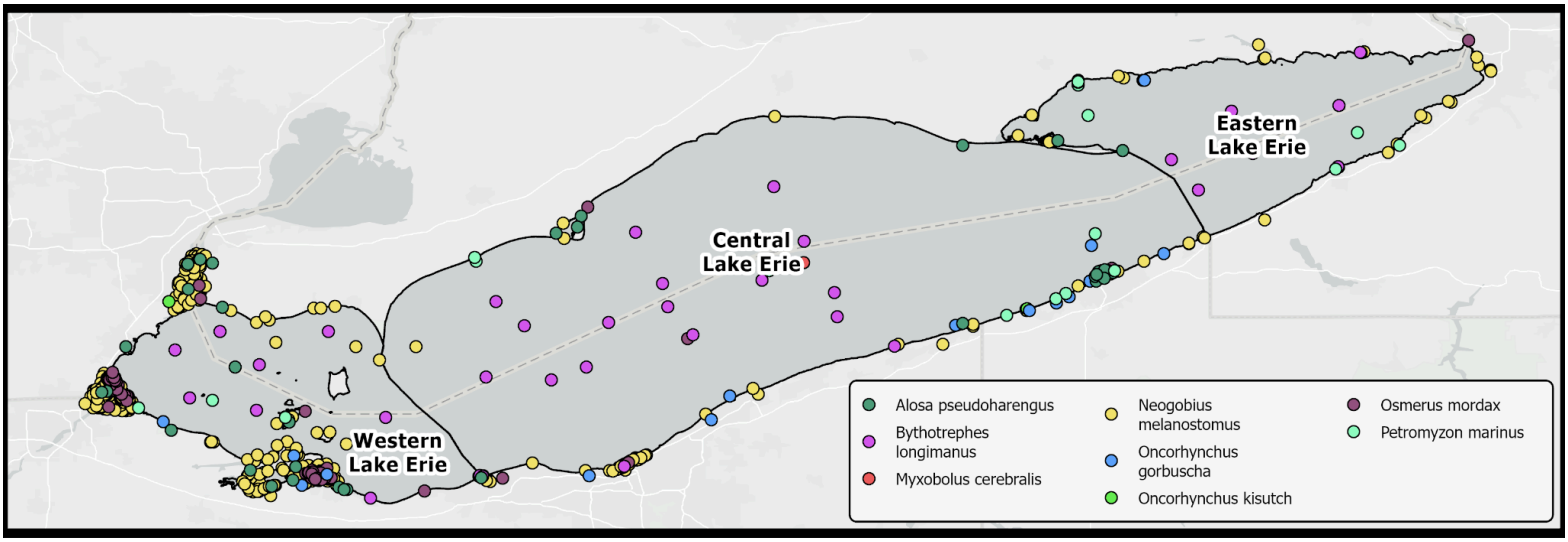


Figure 2. Distribution data for all AIS of interest (Table 2).

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